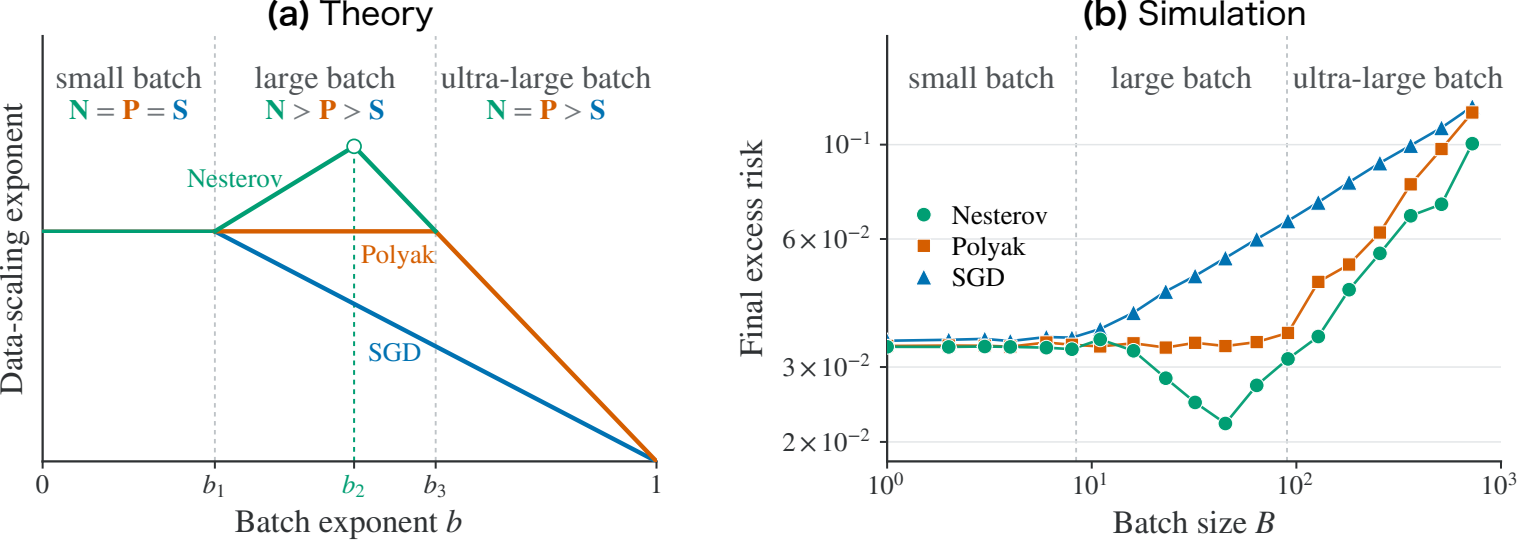
(a) Theory
small batch
N = P = S
large batch
N > P > S
ultra-large batch
N = P > S
Nesterov
Polyak
SGD
Data-scaling exponent
0
b_1
b_2
b_3
1
Batch exponent b
(b) Simulation
small batch
large batch
ultra-large batch
Nesterov
Polyak
SGD
Final excess risk
10^{-1}
6×10^{-2}
3×10^{-2}
2×10^{-2}
10^0
10^1
10^2
10^3
Batch size B